

APPROVED METHOD:
**LIFE TESTING OF SINGLE-BASED
FLUORESCENT LAMPS**
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-65-20(R2023)

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has been approved by the IES.
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should be directed to the IES..

**Prepared by
The IES Testing Procedures Committee**



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CONTENTS

Foreword	1
1.0 Introduction and Scope	1
1.1 Introduction.....	1
1.2 Scope	2
2.0 Normative References	2
2.1 ANSI/IES LS-1-20.....	2
3.0 Nomenclature and Definitions	2
3.1 lamp failure	2
3.2 lamp life.....	2
3.3 rated lamp life.....	2
4.0 Ambient and Physical Conditions	2
4.1 General	2
4.2 Vibration	2
4.3 Temperature	2
4.4 Airflow	3
4.5 Operating Orientation and Lamp Spacing	3
5.0 Electrical Conditions	3
5.1 Power Supply Requirements	3
5.1.1 Input Voltage	3
5.1.2 Input Voltage Wave Shape.....	3
5.1.3 Voltage Regulation.....	3
5.2 Auxiliary Equipment	3
5.2.1 Integrated Ballast	3
5.2.2 Non-integrated Ballast	3
5.2.3 Starters	3
5.3 Electrical Instrumentation.....	3
5.4 Wiring.....	3
5.5 Lamp Holders and Connections	3

6.0 Lamp Test Procedures 4

6.1 Lamp Selection..... 4

6.2 Lamp Handling..... 4

6.3 Lamp Marking..... 4

6.4 Operating Cycle 4

6.5 Recording Failures 4

7.0 Test Report 4

Annex A – Determining Lamp Lumen Maintenance 5

Informative References 5

Foreword

This approved method is a revision of *IES LM-65-14, Approved Method for Life Testing of Compact Fluorescent Lamps*. References were updated.

1.0 Introduction and Scope

1.1 Introduction

The fluorescent lamp is an electric discharge source in which light is produced predominantly by fluorescent powders activated by ultraviolet energy generated by mercury atoms brought to an excited state by collision with electrons in a low-pressure rare-gas-mercury discharge or arc. There are two main ways to generate the electrical discharge. The most common is to pass a current through the gas via electrodes (cathodes) that are connected by wires through the glass envelope. The other way is to induce a current without the need of electrodes by use of high-frequency magnetic induction. This type of discharge source is commonly called an *electrodeless discharge lamp*.

Like most electric discharge lamps, both types of fluorescent lamps require operation in series with a current-limiting device. This device, commonly called a *ballast*, which may be either electromagnetic or electronic, limits the discharge current to the value for which the lamp type is designed. The ballast, in conjunction with the appropriate power supply, provides the required starting capability and maintains the operating lamp voltage and current as required to meet the design specifications for the lamp.

Fluorescent lamps with electrodes require long discharge lengths to be optimally efficient. Normally, the discharge lamp is a cylindrical tube with phosphor coating on the inside surface and an electrode at each end. A long tubular lamp can be made compact by being folded one or more times or spiraled in a helix in such a way that both electrodes are configured to have one connection, hence single-based construction. Further, lamps of any of these bent-tube geometries can be covered with visible-radiation transmitting envelopes

having bulb shapes such as globe (G shaped), tubular (T bulbs), reflector (R and PAR shaped) and others.

Induction-driven electrodeless fluorescent lamps, on the other hand, are necessarily more compact, as the discharge current is required to form a closed loop inside the structure. Electrodeless fluorescent lamps can be globular, A-line types, ring structures, or reflector lamps. Regardless of the electrodeless lamp's outer shape, it has a more convoluted topological construction, as it has to incorporate an electromagnet outside the container for the inert gas and mercury (the gas-discharge volume) around which the current circulates. Two methods are in current use: a tubular coil inserted on the axis of a globular, A-line type, or reflector lamp, or a toroidal coil through which a circular glass tube is inserted.

Finally, there are two categories of compact fluorescent lamps (CFLs): integrated and non-integrated. Non-integrated CFLs, normally only those with electrodes, are composed of a discharge tube, lamp base, and electrical connectors that interface to a lamp holder that connects the external ballast to the lamp. For non-integrated CFLs the ballast is essentially the same as for double-based (linear) fluorescent lamps, and the starting circuits can be the same, i.e., preheat, instant start, rapid start, or program start. Integrated CFLs and electrodeless lamps combine the discharge tube or induction structure, ballast (normally electronic), lamp base, and appropriate housings into one structure. Because the electromagnet, which is part of the ballast, should be in close proximity to the discharge chamber, most electrodeless lamps are integrated (self-ballasted) lamps. Typically, integrated lamps can be directly connected to the branch circuit using standard lamp holders.

It is often important to know the light output, efficacy, and lamp lumen maintenance as well as the life of these lamps. **Annex A** discusses these issues. (For information on the photometry of single-based fluorescent lamps, see ANSI/IES LM-66-20.⁵)

Life performance of single-based fluorescent lamps is typically affected by variables such as operating cycle, vibration, airflow, conditions imposed by auxiliary equipment, and fixture ambient temperature and

orientation. Test conditions and programs should be designed to give comparable results when adopted by various laboratories. The recommendations of this Approved Method have been made with these objectives in mind.

1.2 Scope

This approved method describes the procedures to be followed and the precautions to be observed in order to obtain uniform and reproducible life test measurements of single-based compact fluorescent lamps and integrated electrodeless fluorescent lamps under standard conditions when operating on alternating current (AC) circuits. This document does not include linear, circular-shaped, or single-based U-bent fluorescent lamps.²

2.0 Normative References

2.1 ANSI/IES LS-1-20

Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illuminating Engineering, New York: IES; 2020. Online: <https://www.ies.org/standards/definitions/>. (Accessed 2019 Mar 20).

3.0 Nomenclature and Definitions

The units of electrical measurement used in this test method are the volt, the ampere, and the watt. The units of photometric measurement are the lumen and the candela.

3.1 lamp failure

Lamp failure for this test method refers to the point where a lamp stops operating. Other lamp failures, such as manufacturing defects, are reported but not included in the calculation of lamp life.

3.2 lamp life

Life of a test lamp, for this test method, refers to the actual operating time of the lamp until failure, expressed in hours, and shall not include any off time.

3.3 rated lamp life

Rated lamp life, for this test method, is the life value assigned to a particular type of lamp. This is commonly a statistically determined estimate of median operational life. For life rating, the applicable definition of *median* is “the total operating time at which, under normal operating conditions, 50% of any large group of initially installed lamps is expected to still be operating.”³

4.0 Ambient and Physical Conditions

4.1 General

It is good laboratory practice that the storage and testing of lamps should be undertaken in a relatively clean environment.

4.2 Vibration

Lamps should not be subjected to excessive vibration or shock during life testing.

4.3 Temperature

Ambient (room) temperature shall be controlled between 15 °C (60 °F) and 40 °C (105 °F). When the testing temperature range is exceeded, life testing shall be suspended.

Temperatures above 40 °C (105 °F) may be deleterious to ballasts and components and can affect lamp starting.

For non-integrated lamps, the ballast case temperature at the test point specified by the manufacturer shall be less than or equal to the maximum “allowed” case temperature.⁴

Life testing at elevated temperatures—e.g., reflector type lamps in enclosed fixtures or specially designed elevated-temperature apparatus—may be done. In these cases, the room ambient temperature requirements above still apply, and the test conditions shall be included in the test report.

When specified by the manufacturer, critical lamp temperatures, such as the bulb wall temperature or the control point temperature, shall be maintained.

4.4 Airflow

Airflow should be minimized for proper lamp starting and operation.

4.5 Operating Orientation and Lamp Spacing

The operating position of the lamps under test should be as position is specified, the lamp should be tested in the position that will be used in the application and shall be reported in the test report.

Fixtures used in applications can influence lamp life. If a simulated fixture is used in life testing to approximate this effect, the fixture shall be described in the life test report.

The lamps or fixtures shall be spaced to allow airflow around each lamp or fixture. This is facilitated by designing open life testing racks with minimal structural components that can block airflow.

5.0 Electrical Conditions

5.1 Power Supply Requirements

5.1.1 Input Voltage. Input voltage shall conform to the rated (root-mean-square, RMS) input voltage and frequency of the ballast or integrated lamp. If the rated input voltage is a range, then a nominal value should be selected and shall be reported as a test condition.⁵

5.1.2 Input Voltage Wave Shape. The power supply shall have a sinusoidal voltage wave shape such that the total harmonic distortion does not exceed 3% of the fundamental frequency when operating a purely resistive load.

5.1.3 Voltage Regulation. The input voltage shall be monitored and regulated to within $\pm 2\%$ of the rated RMS value for the duration of the test.

5.2 Auxiliary Equipment

5.2.1 Integrated Ballast. All single-based fluorescent lamps with integrated auxiliary equipment shall be tested as complete units. This applies to both types of lamps, i.e., compact and electrodeless.⁶

5.2.2 Non-integrated Ballast. For single-based compact fluorescent lamps without an integrated auxiliary ballast, the type of ballast chosen may affect lamp life and electrical values during life. In general, the types of ballasts having the widest use should be chosen for the test. Ballasts used in life tests shall comply with the specifications in ANSI C82.1⁷ or ANSI C82.11.⁸ If multiple-lamp ballasts are used, the number of lamps required by the ballast design shall also be used.

5.2.3 Starters. If required, starters shall comply with the specifications in ANSI C78.180.⁹

5.3 Electrical Instrumentation

Accurate recording of elapsed operating time is important in lamp life testing. If used, an elapsed time meter shall accumulate time only when the associated lamps are operating. Video monitoring, current monitoring, or other means may be used to determine elapsed operating time if designed to provide sufficient temporal accuracy.

To facilitate the periodic checking of non-integrated auxiliary equipment or lamp characteristics, appropriate test receptacles may be provided to permit instruments to be connected to each circuit. Instruments shall measure true RMS values.

5.4 Wiring

Wiring of life test racks shall be in accordance with national, state or provincial, and local electrical codes, with particular attention paid to the color coding on the input wiring to the ballast. The ballast shall be grounded per the manufacturer's specifications.

5.5 Lamp Holders and Connections

Lamp holders shall be rated for the specific test application under which they are used.

During life testing of preheat and rapid-start non-integrated lamps, it is important that the orientation of the pins in relation to the ballast leads be marked. This orientation shall be used throughout all testing, including photometric and electrical measurements.¹

Inspection of lamp holder contacts and ballast verification should be made whenever lamps are placed on the life test rack.

6.0 Lamp Test Procedures

6.1 Lamp Selection

The lamps shall be selected to represent the purpose of the test. However, if the test is designed to evaluate specific characteristics, the sample selection needs to take this into consideration. The value of the test will depend upon the method of sampling, size of the sample, conditions of testing, and other factors.¹⁰

6.2 Lamp Handling

Prior to operation, lamps shall be cleaned to eliminate surface contamination.

If the lamps are to be removed from the life test racks for testing, evaluation, transporting, and storing, maintaining the lamps in the same orientation as during operation on the life test racks can reduce the lamp stabilization time for evaluation and for subsequent continuation of life testing. This practice can be effective and is recommended for lamp types that have defined control points or cold chambers.

6.3 Lamp Marking

Individual lamps shall be tracked during life testing. Lamps can be identified by markings applied directly to the lamps, or by labels that can be attached to the lamps during transport and evaluation or to the life test rack position occupied by the lamp during the life test. The identification method selected should take into account the effect of exposure to light, UV and IR radiation, and heat.

Suitable marking methods or materials include durable bar coding, ceramic ink marking, high-temperature markers, or other methods that endure or can be periodically renewed for the duration of the life testing.

6.4 Operating Cycle

The standard life test operating cycle shall be 180 minutes on, 20 minutes off. Life data from different operating cycles cannot meaningfully be compared without additional information. There is no industry consensus method for accelerated life testing of single-based fluorescent lamps.

6.5 Recording Failures

Checking for lamp failures, either by visual observation or automatic monitoring, shall be done at an interval of no more than 1% of the rated life.

The recorded failure time shall be determined as the midpoint of the last monitored interval. Each failure shall be investigated to make certain that it is actually a lamp failure and is not caused by improper functioning of the auxiliary equipment or lamp holders. If multiple-lamp ballasts are used, the failed lamp shall be replaced by another operating lamp of the same lamp type.

Lamp failures can be divided roughly into three classes: those caused by manufacturing defects, those caused by normal failures, and lamps showing markedly low light output.

7.0 Test Report

Test reports shall include the hours at which lamps failed, including seasoning time for each lamp installed at the start of the test.

The report shall also list all pertinent data concerning conditions of testing, type of equipment, and lamp types being tested. Typical items reported are:

- Date and testing agency
- Manufacturer's name and lamp designation
- Type of test
- Number of lamps tested
- Description of lamps
- Description of auxiliary equipment
- Operating cycle
- Ambient temperature
- Lamp position during test
- Input voltage (RMS value and frequency)
- Lamp monitoring interval
- Times at which lamps failed (hours)
- Classification of failures
- Median of lamp test when completed

Annex A – Determining Lamp Lumen Maintenance

It is often desirable to measure the lamp lumen or light output maintenance during the course of a life test. Photometric measurements are made, at a minimum,

after 1,000 hours and at 40% of rated life. These values are compared to the initial lamp output, measured after the initial seasoning time of the lamp. Historically, regulatory agencies have required measurements at these two points. More intervals can be used to further define and understand the shape of the lamp lumen maintenance curve.

INFORMATIVE REFERENCES

- 1 Illuminating Engineering Society. ANSI/IES LM-66-20, Approved Method: Electrical and Photometric Measurements of Single-Based Fluorescent Lamps, New York: IES; 2020.
- 2 Illuminating Engineering Society. ANSI/IES LM-40-20, Approved Method: Life Testing of Fluorescent Lamps. New York: IES; 2020.
- 3 Illuminating Engineering Society. ANSI/IES LP-4-20, Lighting Practice: Electric Light Sources – Properties, Selection, and Specification. New York: IES; 2020.
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- 10 National Institute of Standards and Technology. NIST/SEMATECH e-Handbook of Statistical Methods. Washington, DC: NIST. Online: <http://www.itl.nist.gov/div898/handbook/>. (Accessed 2019 Mar 21).



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